

House Price Prediction – Project Summary

Objective:

The objective of this project was to build a machine learning model to predict house prices using property features such as area, bedrooms, bathrooms, parking, furnishing status, and location-related attributes. The project also aimed to identify the factors that most influence house prices.

Dataset Description:

The Housing Prices Dataset contains 545 records and 13 columns. The target variable is price, while the remaining columns were used as features. The dataset contained no missing values, and categorical variables were converted into numerical form using one-hot encoding.

Model Performance

The dataset was split into training and testing sets using an 80:20 ratio.

Linear Regression MAE = 970,043 RMSE = 1,324,507 R^2 Score = 0.653

Random Forest Regressor MAE = 1,006,663 RMSE = 1,383,659 R^2 Score = 0.621

Linear Regression performed better than Random Forest, achieving lower prediction errors and a higher R^2 score.

Key Findings

The analysis showed that house price is influenced most by property area, number of bathrooms, number of bedrooms, parking availability, air conditioning, and preferred location. Larger houses with better amenities generally command higher prices. The correlation analysis and feature importance results indicated that area has the strongest relationship with house price.

The Linear Regression model achieved an R^2 score of approximately 0.65, which means that the model can explain about 65% of the variation in house prices. This indicates a reasonably good predictive capability for a simple regression model.

One interesting observation was that location-related features and property amenities had a noticeable impact on house prices in addition to physical property size.

Business Recommendation

Real estate companies should prioritize larger properties located in preferred residential areas and emphasize features such as air conditioning, parking facilities, and multiple bathrooms during marketing campaigns. These characteristics have the greatest influence on house prices and can help maximize property value and sales revenue.